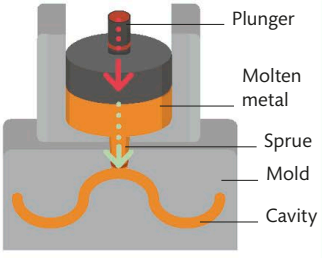
Working with metals

Most metals are produced as simple ingots, sheets, or bars, which usually require shaping or joining to other items to make finished products. Metals may also need to be treated to improve their properties, for example, to make them easier to shape or more corrosion resistant.

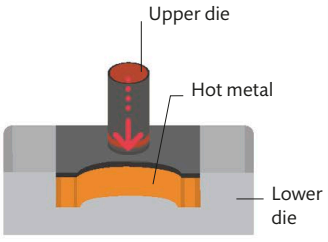
Shaping metals

Metals have a crystalline structure that breaks down when heat is applied. The metal softens and then becomes molten, allowing it to be easily shaped. As the metal cools, it recrystallizes and becomes hard once more. Processes that take advantage of these transformations to shape metals are known as hot working and include casting, extruding, forging, and rolling. Metals can also be worked without the application of heat, by processes known as cold working. In these processes, change in the metal is brought about through mechanical stresses rather than heat.

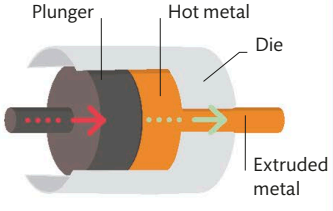
HOT METHODS



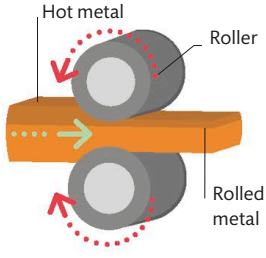
Casting
Molten metal is poured through a channel (called a sprue) into a mold. Once the metal has cooled, it can be extracted. Casting is typically used for complex 3-D shapes.



Forging
Forging replaces the blacksmith's hammer and anvil with modern machinery. The hot metal is compressed to the desired shape between two shaped dies—one die is fixed, and one is movable.



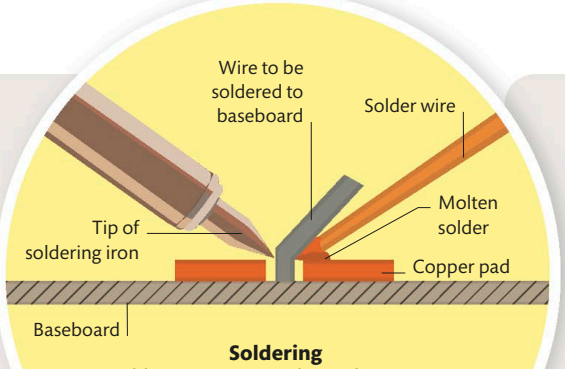
Extruding
The metal is softened with heat and then pushed through a die. Extrusion is used to create uniform cross sections, typically simple shapes such as rods or pipes.



Rolling
In this process, a slab of hot metal is fed through rollers to reduce its thickness. Rolling is used to create sheet metal and other structural components.

Joining metals

The main methods for joining metals are soldering, welding, and riveting. Soldering and welding rely on the principle that metals become molten and pliable when heated and then return to a hardened state on cooling. Soldering forms the weakest bond, because it uses a soft metal with a lower melting point to act as the "glue." In welding, the two metals to be joined are melted and fused together, creating an exceptionally strong bond. Riveting also creates a strong bond and has a higher tolerance for thermal expansion and contraction. It is also cheaper than welding. However, riveting is less aesthetically pleasing than welding and so is usually used on internal or industrial structures.



Soldering
Soldering is commonly used to create bonds in electronic equipment. Soft solder is melted so that it flows into the space between two pieces of metal, joining them together when it cools.